

KARTIK MARATHE

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SUMMARY

Master of Science in Computer Science candidate at Arizona State University with two years of full stack experience. Skilled in Spring Boot microservices, React front ends, AWS deployments, and building ETL pipelines for data engineering workflows.

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, JavaScript, SQL, Bash, TypeScript, Go
- **Cloud & Databases:** AWS, Azure, Docker, Kubernetes, PostgreSQL, MySQL, MongoDB
- **Development:** Spring Boot, React.js, Node.js, Next.js, Angular, HTML, CSS, REST APIs, GraphQL, JUnit, Mockito, Tailwind CSS
- **Tools:** Spark, Kafka, Redis, Jenkins, GitLab CI/CD, Git, GitHub, Jira, Postman, Agile, Trello

EXPERIENCE

Research Aide (Volunteer) | Arizona State University

Sep 2024 - Present

- Explored LLM-based techniques for surfacing latent code smells and design anti-patterns, crafting structured prompts grounded in refactoring patterns and cognitive load theory to generate concrete remediation suggestions.
- Performed AST-based analysis to rank high-impact files for technical-debt cleanup in large open-source codebases.

Associate Software Engineer | The Boeing Company

Aug 2022 - Jul 2024

- Migrated legacy desktop app to a **Spring Boot** web platform; decommissioned legacy servers, enabled automated deployments, and **saved ~\$72k/year** in infra costs.
- Designed an **Apache Solr 8** search stack for **10k+ PDF** records: PDF-to-JSON parsing, REST ingestion, fuzzy & phrase search with faceting, integrated with an **Angular 14** UI for rapid querying.
- Orchestrated **GitLab CI/CD** (staged builds, unit tests, env-specific deploys) for Java + Angular services, **reducing release time from ~60 min to ~10 min** and eliminating manual error-prone steps.
- Built and maintained **50+ REST endpoints** across **4 Spring Boot microservices**, enforced controller/service/repository tests achieving **~90% line coverage** with **JUnit 5** and **Mockito**.
- Expanded content management system with **5 REST APIs**, shrinking the publish flow from **6 to 3 steps** and **7 to 3 minutes**.

Research Intern | Ikshana (Indian Institute of Science)

Jan 2022 – Jul 2022

- Scoped a Python (Pandas) ETL approach to clean and time-align **10+ clinical and sensor streams**; documented schema, resampling, and feature plan to support urinary incontinence analysis with scikit-learn.
- Developed an Android Studio (Java) app with **Bluetooth Low Energy** (BLE/GATT) to scan, pair, and remotely control devices such as read, write and notify using a **Material 3** UI.

PROJECTS

Wealth Wizard | <https://ai-finance-tracker-seven.vercel.app/>

- Engineered a comprehensive finance platform using Next.js, PostgreSQL, Inngest, and Arcjet, featuring multi-account support, recurring transactions, real-time expense tracking, budget alerts, and interactive reporting dashboards.
- Enabled users to **set budgets** and send notifications when approaching their limits, along with **monthly spending insights**.
- Structured **financial dashboards** with interactive visualizations for expense tracking and **account statistics**.

Spotify with Chat | <https://spotify-clone-gi9v.onrender.com/>

- Launched an advanced Spotify-inspired web application leveraging the MERN stack, incorporating comprehensive admin capabilities to manage a library of **100+ songs** and **20+ albums** efficiently.
- Incorporated real-time chat functionality and user status tracking using **Socket.io**, enhancing interactive communication.
- Deployed on **AWS EC2** behind an Application Load Balancer with **Auto Scaling Groups**, and stored song assets in **AWS S3**.

Energy Visualization

- Collaborated in a 4-member team to develop a full-stack web application using Spring Boot, React, and D3.js, visualizing energy import/export and renewable energy generation data for **200+ countries** with a 3D globe rendered via Three.js.
- Programmed scroll-synced storytelling that animates **60+ years** of data in linked line charts for temporal trend exploration.
- Built an **animated** country-centric network graph, where nodes are renewable capacity and links are import-export flows.

EDUCATION

Master of Science, Computer Science

Aug 2024 - May 2026

ARIZONA STATE UNIVERSITY, Tempe, AZ (GPA: 4.0/4.0)

- **Coursework:** Data Mining, Cloud Computing, Mobile Computing, Data Visualization

Bachelor of Engineering, Electronics and Communication

Aug 2018 - Jul 2022

R.V. COLLEGE OF ENGINEERING, Bengaluru, India (GPA: 3.47/4.0)

- **Coursework:** Database Management Systems, Intelligent Systems, Data Structures and Algorithms, Web Programming

AWARDS & RESEARCH

- **Winner**, Innovation Hacks 2025 (ASU) for Resume customization & compatibility scoring app.
- **Winner**, TIAA Fund the Future Spark Challenge 2025 for creating Optimized retirement contribution matching app
- **Conference Presentation**, IEEE INDICON for VLSI Floorplan Optimization Tool.